Data Dictionary

AdventureWorks Power BI Report - Tables, Columns & Measures

This dictionary lists every table and column used in the model, and every DAX measure with its formula and a plain-English description. Columns marked "(hidden)" are hidden from Report view but retained in the model for relationships or calculations. The Returns table is simulated (see its description).

Tables & Columns

Sales

Fact table. One row = one sales order line.

Column	Data Type	Meaning
Customer Key	Whole Number (hidden)	Foreign key linking to Customer.
Due Date Key	Whole Number (hidden)	Foreign key linking to Calendar; the promised delivery date.
Extended Amount	Currency (hidden)	Line-level sales amount before adjustments.
Order Date Key	Whole Number (hidden)	Foreign key linking to Calendar; the date the order was placed.
Order Quantity	Whole Number (hidden)	Number of units sold on this order line.
Product Key	Whole Number (hidden)	Foreign key linking to Product.
Product Standard Cost	Currency (hidden)	Standard cost of the product at time of sale.
Reseller Key	Whole Number (hidden)	Foreign key linking to Reseller.
Sales Amount	Currency	Final revenue amount for this order line. Used by Total Sales and most revenue measures.
Sales Order Line Key	Whole Number (hidden)	Unique identifier for the order line; the table's grain.
Sales Territory Key	Whole Number (hidden)	Foreign key linking to SalesTerritory.
Ship Date Key	Whole Number (hidden)	Foreign key linking to Calendar; the date the order shipped.
Total Product Cost	Currency	Total cost of goods for this order line. Used by Total Cost.
Unit Price	Currency	Price per unit before discount.
Unit Price Discount Pct	Decimal (%)	Discount applied to the unit price, as a percentage.

Product

Dimension table. One row = one product.

Column	Data Type	Meaning
Category	Text	Top-level product grouping (e.g. Bikes, Accessories).
Color	Text	Product color.
List Price	Currency	Standard retail price.
Model	Text	Product model name.
Product Key	Whole Number (hidden)	Unique identifier for the product; the table's grain.
Product Name	Text	Full display name of the product. Also the field the Product Detail drill-through page is based on.
SKU	Text	Stock keeping unit code.
Standard Cost	Currency	Standard manufacturing/acquisition cost.
Subcategory	Text	Second-level product grouping under Category.

Customer

Dimension table. One row = one customer.

Column	Data Type	Meaning
City	Text	Customer's city.
Country-Region	Text	Customer's country or region.
Customer ID	Text (hidden)	Business-facing customer identifier.
Customer Key	Whole Number (hidden)	Unique identifier for the customer; the table's grain.
Customer Name	Text	Full customer name.
Customer Segment	Text (calculated column)	Gold / Silver / Bronze / No Sales tier, based on the customer's Total Sales. See the Measures section for the formula and why it's a column, not a measure.
Postal Code	Text	Customer's postal code.
State-Province	Text	Customer's state or province.

Reseller

Dimension table. One row = one reseller.

Column	Data Type	Meaning
Business Type	Text	Type of reseller business.
City	Text	Reseller's city.
Country-Region	Text	Reseller's country or region.
Postal Code	Text	Reseller's postal code.
Reseller ID	Text (hidden)	Business-facing reseller identifier.
Reseller Key	Whole Number (hidden)	Unique identifier for the reseller; the table's grain.
Reseller Name	Text	Full reseller name.
State-Province	Text	Reseller's state or province.

SalesTerritory

Dimension table. One row = one sales territory.

Column	Data Type	Meaning
Country	Text	Country the territory belongs to.
Group	Text	Broad territory group (e.g. North America, Europe, Pacific). Used for the static RLS roles.
Region	Text	Specific region/territory name (e.g. Southwest, Northeast).
Territory Key	Whole Number (hidden)	Unique identifier for the territory; the table's grain.

Calendar

Date dimension, generated with List.Dates() in Power Query. Marked as a Date Table for time intelligence. One row = one calendar date.

Column	Data Type	Meaning
Date	Date	The calendar date; the table's grain.
Date Key	Whole Number (hidden)	Numeric surrogate key matching Sales' date foreign keys.
Day of Week	Text	Name of the weekday.
Is Weekend	Text/Boolean	Flags whether the date falls on a weekend.
Month Name	Text	Full month name.
Month Number	Whole Number	Numeric month (1-12); used to sort Month Name correctly.
Quarter	Whole Number	Calendar quarter.
Week Number	Whole Number	ISO week number.
Year	Whole Number	Calendar year.

Budget

Standalone table (no formal relationship to Product) used to demonstrate TREATAS cross-table filtering.

Column	Data Type	Meaning
Budget Amount	Whole Number/Currency	Budgeted sales target for the category.
Category	Text	Product category the budget applies to. Matches Product[Category] by value only, not by a formal relationship.

Employees

Standalone table (Enter Data), used only for Dynamic RLS.

Column	Data Type	Meaning
Name	Text	Employee name, matched against USERPRINCIPALNAME() for dynamic RLS.
Territory	Text	Territory the employee is restricted to; matched against SalesTerritory[Group].

Returns

Simulated table, built in Power Query because AdventureWorks has no real returns data. A random 8% of Sales rows are treated as returned. One row = one returned sales order line. Linked to Sales on Sales Order Line Key (one-to-one). The values re-randomise each time the data refreshes.

Column	Data Type	Meaning
Days To Return	Whole Number	Simulated number of days between the order and the return (3 to 30).
Return Date	Date	Simulated date of the return (order date plus Days To Return).
Return Month	Date (calculated column)	First day of the month of the return. Used for the monthly returns trend chart.
Return Quantity	Whole Number	Simulated units returned on that order line (never more than the quantity ordered).
Return Reason	Text	Randomly assigned reason: Defective Product, Wrong Item Shipped, Changed Mind, Not as Described, or Damaged in Transit.
Sales Order Line Key	Whole Number	Links to Sales Order Line Key in Sales; the table's grain.

Measures

All measures live in the dedicated Measures table. 32 measures are documented below, covering foundational aggregations, time intelligence, error handling, TREATAS cross-table filtering, ranking/segmentation, CALCULATE modifiers, and the simulated Returns measures.

Measure	DAX Formula	Description
Total Sales	`SUM(Sales[Sales Amount])`	Total revenue from all sales transactions.
Total Cost	`SUM(Sales[Total Product Cost])`	Total cost of goods sold.
Gross Profit	`[Total Sales] - [Total Cost]`	Revenue minus cost of goods sold.

Measure	DAX Formula	Description
Gross Profit Margin %	`VAR TotalSalesAmt = [Total Sales] VAR TotalGrossProfit = [Gross Profit] RETURN DIVIDE(TotalGrossProfit, TotalSalesAmt, 0)`	Gross profit as a percentage of total sales.
Total Customers	`DISTINCTCOUNT(Sales[Customer Key])`	Count of unique customers who made a purchase.
Total Units Sold	`SUM(Sales[Order Quantity])`	Total quantity of items sold.
Total Orders	`DISTINCTCOUNT(Sales[Sales Order Line Key])`	Count of distinct order lines.
Average Order Value	`VAR TotalSalesAmt = [Total Sales] VAR TotalOrderCount = [Total Orders] RETURN DIVIDE(TotalSalesAmt, TotalOrderCount, 0)`	Average revenue per order.
Sales YTD	`CALCULATE(TOTALYTD([Total Sales], Calendar[Date]), REMOVEFILTERS(Calendar[Year]))`	Cumulative sales from Jan 1 to the latest date in context.
Sales PY	`CALCULATE([Total Sales], SAMEPERIODLASTYEAR(Calendar[Date]))`	Sales for the same period, one year earlier.
Sales YoY Change $	`VAR CurrentSales = [Sales YTD] VAR PriorSales = [Sales PY] RETURN CurrentSales - PriorSales`	Absolute change in sales vs. the same period last year.
Sales YoY Change %	`VAR CurrentSales = [Sales YTD] VAR PriorSales = [Sales PY] RETURN DIVIDE(CurrentSales - PriorSales, PriorSales, 0)`	Percentage change in sales vs. the same period last year.
Sales MTD	`TOTALMTD([Total Sales], Calendar[Date])`	Cumulative sales, month-to-date.
Sales QTD	`TOTALQTD([Total Sales], Calendar[Date])`	Cumulative sales, quarter-to-date.
Rolling 3-Month Sales	`VAR MaxSaleDate = MAX(Calendar[Date]) VAR RollingDates = DATESINPERIOD(Calendar[Date], MaxSaleDate, -3, MONTH) RETURN CALCULATE([Total Sales], RollingDates)`	Trailing 3-month sales total.
Rolling 12-Month Sales	`VAR MaxSaleDate = MAX(Calendar[Date]) VAR RollingDates = DATESINPERIOD(Calendar[Date], MaxSaleDate, -12, MONTH) RETURN CALCULATE([Total Sales], RollingDates)`	Trailing 12-month sales total.
Has Sales	`VAR CustomerSales = CALCULATE(COUNTROWS(RELATEDTABLE(Sales)), KEEPFILTERS(ALL(Sales[Order Quantity]))) RETURN CustomerSales > 0`	Error-handling flag: TRUE/FALSE for whether the current filter context has any related sales rows.
Budget Amount by Category	`CALCULATE(SUM(Budget[Budget Amount]), ALL(Budget), TREATAS(VALUES(Product[Category]), Budget[Category]))`	Pulls the budget amount from the unrelated Budget table by matching Category via TREATAS.
Budget vs Actual Variance	`VAR ActualSales = [Total Sales] VAR BudgetTarget = [Budget Amount by Category] RETURN ActualSales - BudgetTarget`	Actual sales minus the budgeted amount for the category.
Budget Attainment %	`DIVIDE([Total Sales], [Budget Amount by Category], 0)`	Actual sales as a percentage of budget.
Top 10 Products by Sales	`IF(RANKX(ALL(Product[Product Name]), [Total Sales]) <= 10, [Total Sales])`	Shows the sales value only for the top 10 ranked products; blank otherwise.
Product Rank within Category	`RANKX(FILTER(ALL(Product), Product[Category] = MAX(Product[Category])), [Total Sales])`	A product's sales rank within its own category.
High Value Customer	`IF([Total Sales] > 50000, "Yes", "No")`	Flags a customer as high value if their sales exceed the threshold.

Measure	DAX Formula	Description
High Value Customer Count	`IF(ISBLANK(CALCULATE(DISTINCTCOUNT(Customer[Customer Key]), Customer[Customer Segment] IN {"Silver", "Gold"}, Customer[Customer Name] <> "[Not Applicable]")), 0, CALCULATE(DISTINCTCOUNT(Customer[Customer Key]), Customer[Customer Segment] IN {"Silver", "Gold"}, Customer[Customer Name] <> "[Not Applicable]"))`	Count of real (non-"[Not Applicable]") customers in the Silver or Gold segments. Wrapped in an ISBLANK check so it displays as 0 rather than blank when no customer qualifies.
Customer Segment (Legacy)	`VAR CustSales = [Total Sales] RETURN SWITCH( TRUE(), CustSales >= 100000, "Gold", CustSales >= 50000, "Silver", CustSales > 0, "Bronze", "No Sales")`	Original measure-based version of the customer segment tier. Superseded by the Customer[Customer Segment] calculated column (which can be placed on a chart axis); kept and renamed to avoid a naming clash, not used in the live report.
Days Since Last Purchase	`DATEDIFF( MAXX(RELATEDTABLE(Sales), RELATED(Calendar[Date])), TODAY(), DAY)`	Days elapsed since a customer's most recent order, calculated via the Calendar date rather than the raw numeric date key.
Pareto %	`VAR RunningTotal = CALCULATE([Total Sales], FILTER(ALL(Product[Product Name]), RANKX(ALL(Product[Product Name]), [Total Sales]) <= RANKX(ALL(Product[Product Name]), [Total Sales], CALCULATE([Total Sales], REMOVEFILTERS())))) VAR GrandTotal = CALCULATE([Total Sales], ALL(Product)) RETURN DIVIDE(RunningTotal, GrandTotal, 0)`	Cumulative % of total sales contributed by products at or above the current product's rank (80/20 Pareto analysis).
No Data Message	`IF(ISBLANK([Total Sales]), "No data available for the selected filters", BLANK())`	Feeds a Card visual on Sales Analysis; displays an empty-state message only when Total Sales is blank for the current filters.
Total Returns	`COUNTROWS(Returns)`	Count of returned order lines (simulated data).
Return Rate %	`DIVIDE([Total Returns], [Total Orders], 0)`	Returned order lines as a share of all orders (simulated, about 8%).
Return Value Lost	`SUMX(Returns, RELATED(Sales[Sales Amount]))`	Sales amount of the returned order lines. Counts the full line value, not prorated by quantity returned.
Average Days to Return	`AVERAGE(Returns[Days To Return])`	Average gap in days between the order and the return (simulated).